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Transformer Normalisation Layers and the Independence of Semantic Subspaces

Stephen Menary
University of Manchester, UK
stephen.menary@manchester.ac.uk

Samuel Kaski
[1] University of Manchester, UK
[2] Aalto University, Finland
samuel.kaski@aalto.fi

André Freitas
[1] Department of Computer Science, University of Manchester, UK
[2] Idiap Research Institute, Switzerland
[3] National Biomarker Centre, CRUK-MI, University of Manchester, UK
andre.freitas@manchester.ac.uk

Abstract

Recent works have shown that transformers can solve contextual reasoning tasks by internally executing computational graphs called *circuits*. Circuits often use attention to logically match information from subspaces of the representation, e.g. using position-in-sequence to identify the previous token. In this work, we consider a *semantic subspace* to be any independent subspace of the latent representation that can fully determine an attention distribution. We show that `Pre-Norm`, the placement of normalisation layer used by state-of-the-art transformers, violates this ability unless the model learns a strict representation structure of orthogonal spheres. This is because it causes linear subspaces to interfere through their common normalisation factor. Theoretically, we analyse circuit stability by modelling this interference as random noise on the L_2-norms of the query/key/value vectors, predicting a phenomenon of *circuit collapse* when sparse-attention shifts to a different token. Empirically, we investigate the sensitivity of real-world models trained for mathematical addition, observing a 1% rate of circuit collapse when the norms are artificially perturbed by $\lesssim$10%. We contrast `Pre-Norm` with `QKV-Norm`, which places normalisation *after* the attention head's linear operators. Theoretically this relaxes the representational constraints. Empirically we observe comparable in-distribution but worse out-of-distribution performance.

1 Introduction

Transformer-based models [1] are commonplace in machine learning, providing state-of-the-art contextual reasoning in domains ranging from natural language [2, 3] to protein-folding [4–6] and theoretical physics [7]. Recent interpretability work investigates the internal mechanisms that lead to specific model behaviours [8–15]. This is important for predicting behaviour in new environments, enables practitioners to match the inductive bias of a model with the structure of its task, and informs the design of architectures that promote desirable behaviour.

Two such works discovered complete *circuits* [16] in trained transformers [8–10]. These are computational graphs that dominate the model prediction when activated in a specialised context. They perform a type of *algorithmic reasoning* by internally executing a sequence of logical operations, using attention to pass information between memory buffers that begin as token embeddings and become increasingly abstract. Furthermore, a number of attention heads have been identified as

performing logical operations (see [11] section 5). To understand transformer behaviour, an important goal is to understand how logical attention heads operate, and their generality beyond the simple cases that facilitate interpretability.

One key observation is that **the attention distribution is sometimes fully-determined by an independent subspace of the representation** - for example, an attention layer can identify the previous token by accessing a subspace that encodes only position-in-sequence. Indeed, low-rank weight matrices can only access linear subspaces by construction. A second observation is that, like most deep architectures, transformers use normalisation layers to improve training stability. A leading choice is to place normalisation at the *input* to each attention layer, which we call `Pre-Norm` [17]. Some interpretability works ignore this layer because it has a linear-up-to-scale structure, absorbing the linear part into adjacent weights. In this work we argue that the layer is important, because **Pre-Norm causes independent linear subspaces to interfere through a common normalisation factor, preventing their separation by linear attention layers**.

The purpose of this work is to ask: if the use of independent subspaces is generally important, what are the expected consequences of `Pre-Norm` for (i) the latent representation structure, and (ii) circuit stability? To answer this, we take an abstract approach that complements direct interpretability by considering general behaviour beyond the interpretable limit. Our contributions are:

1. **Conceptual:** we identify interference between independent subspaces as a potential destabiliser of circuits caused by `Pre-Norm`. We suggest *separability of latent subspaces* as a target for study, and show it is easily satisfied by the alternative `QKV-Norm`. This differs from `Pre-Norm` by placing the normalisation layer after the linear operators. It is similar to `QK-Norm`, for which sparse evidence currently exists [18–20].
2. **Theoretical:** we formalise a *semantic subspace* as any independent subspace of the latent representation that can fully determine the attention distribution. We show that `Pre-Norm` can only achieve this when semantic subspaces are spherical and mutually orthogonal. By contrast, `QKV-Norm` requires only that subspaces be linearly independent, matching the `No-Norm` case in this sense. We study the stability of attention to subspace interference, predicting a potentially problematic phenomenon of *circuit collapse* when a sparse-attention distribution changes which embedding it attends to.
3. **Experimental:** we measure the sensitivity of trained models to simulated interference in a numerical addition task. Constraining our predictions, we find that (i) `Pre-Norm` models induce a narrower distribution of embedding L_2-norms than `QKV-Norm`, (ii) we bound the spread of L_2-norms to $\pm 20\%$ with 90% coverage, and (iii) the circuit collapse phenomenon occurs at a rate of 1% when norms are perturbed by $\mathcal{O}(10\%)$.

2 The idea

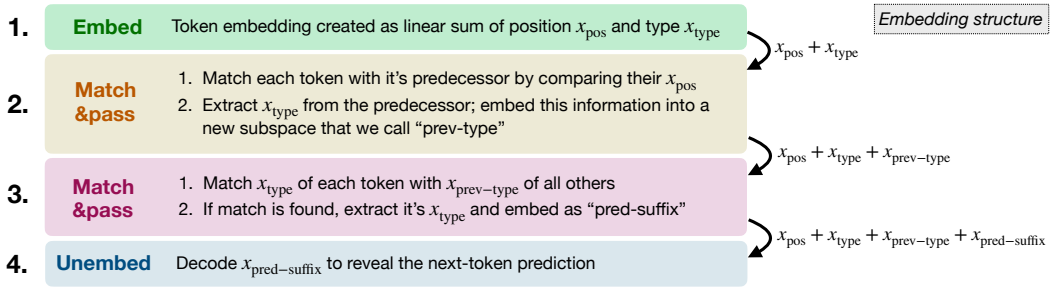
Independent subspaces are observed in real-world transformer circuits

Before providing a formal definition in section 5, we explain what we mean by a *semantic subspace* of the latent representation. To emphasise that this is observed in real-world models, we use a known example: the *induction circuit* [8, 9]. This two-layer circuit emerges in next-token-prediction models and implements a simple contextual reasoning algorithm called *prefix-matching*.

Consider text to be a sequence of tokens[1], and our task is to predict the next token at every point. The induction circuit solves this by copying a previous example from the context window: e.g. if the input includes the phrase "Harry Potter" and the last observed word was "Harry", the induction circuit will predict that "Potter" comes next. This solves the task even if the combination "Harry Potter" never occurred in the training data.

To achieve this, we initially create an embedding for each token, encoding it's *position* and *type*. Attention layers then *copy information between embeddings in a directed way*, using two components that determine (i) *which* embeddings to extract information from, and (ii) *what* to extract. Remarkably, the model learns to implement logical gates that we will call "match&pass", internally composing the algorithm:

[1] In this example, we tokenise per-word to help with visualisation.



Each match&pass step operates only on an independent subspace of information, which we will call a *semantic subspace*. In this example, there are four semantic subspaces corresponding to *position*, *type*, *prev-type*, and *pred-suffix*. We observe that the latent embeddings can contain various information, and it is instructive to think of them as memory buffers rather than tokens. The principle of *composing logical operations that act on latent semantic subspaces* is also observed in the more complex example of indirect-object identification in GPT2-Small [10].

The problem with Pre-Norm

We express the latent embeddings as $x = \sum_\alpha x_\alpha$ where x_α encodes the value of concept α. This is important, because linear-attention layers extract information from x using linear operators (section 4), and can only isolate x_α if each subspace $\{x_\alpha \mid \alpha\}$ is *linearly independent*. In other words, there must always exist a linear projection operator P_α such that $P_\alpha x = x_\alpha$.

Most transformers use either `RMSNorm` [21] or `LayerNorm` [22] for their internal normalisation layers. Geometrically, `RMSNorm` projects a vector $z \in \mathbb{R}^N$ onto the unit-sphere S^{N-1} according to

$$z \;\rightarrow\; \frac{z}{|z|} \qquad\qquad \text{where } |z| \;\triangleq\; \sqrt[+]{\sum_{i=1}^{N} z_i^2} \text{ is the } L_2\text{−norm.} \tag{1}$$

`LayerNorm` is similar, projecting onto the sphere S^{N-2} defined perpendicular to the direction 1^N. This does not affect our analysis, and we focus on `RMSNorm` for simplicity. Normalisation layers sometimes also include gain and/or bias parameters, applying a stretch-and-translate to the sphere. `Pre-Norm` [17] normalises the latent embeddings at the *input* to every attention layer. Consider the example $x = x_{\text{pos}} + x_{\text{type}} + x_{\text{prev-type}}$. Applying `Pre-Norm`, we find $P_{\text{pos}} x = x_{\text{pos}}$ is replaced by

$$P_{\text{pos}} \frac{x}{|x|} \;=\; \frac{P_{\text{pos}} x}{|x|} \;=\; \frac{x_{\text{pos}}}{|x_{\text{pos}} \;+\; x_{\text{type}} \;+\; x_{\text{prev-type}}|} \tag{2}$$

Therefore it is impossible for a linear-attention layer to extract x_{pos} without interference from x_{type} and $x_{\text{prev-type}}$, unless $|x_{\text{pos}} + x_{\text{type}} + x_{\text{prev-type}}|$ is a constant. In general, we have $P_\alpha \frac{x}{|x|} = \frac{x_\alpha}{|\sum_\beta x_\beta|}$, and semantic subspaces are entangled unless $|\sum_\alpha x_\alpha|$ is constant. This is only possible if $|x_\alpha|^2 = const_\alpha \;\forall\; \alpha$, i.e. every subspace is a sphere, and $x_\alpha^T x_\beta = 0 \;\forall\; x_\alpha, x_{\beta \neq \alpha}$, i.e. all spheres are orthogonal (to maintain independence). This has several possible implications:

1. It is a restrictive structure that must be learned during training, with unknown difficulty. Finite steps of gradient descent may separate the model from the manifold of acceptable representations, hindering the learning of circuit components that require semantic separation, like match&pass, especially when training with large learning rates.
2. The constraint $|x_\alpha|^2 = const_\alpha$ removes a degree of freedom for every α, reducing the information capacity of the embedding space. For example, an embedding on $\mathbb{R}^5$ could have the two-subspace structure $\mathcal{S}^2 \bigoplus \mathcal{S}^1$ but not $\mathcal{S}^2 \bigoplus \mathcal{S}^2$.
3. We hypothesise that the structure may be violated by (i) a tradeoff with other representational effects, (ii) imperfect model training, or (iii) encountering unexpected semantic combinations at inference-time when generalising out-of-distribution. These would cause semantic subspaces to interfere through their common normalisation factor, manifesting as noise on the L_2-norms of the {query, key, value} vectors.
4. It is a structure that we can search for empirically.